Practical 2: 8 Puzzle using DFS

```
from collections import deque

start = (
   1, 2, 3,
   4, 5, 6,
   0, 7, 8
)

goal = (
   1, 2, 3,
   4, 5, 6,
   7, 8, 0
)

stack = [(start, [start])]   # (state, path)
visited = [start]

while stack:
   s, path = stack.pop()    # LIFO -> DFS

   if s == goal:
      print("GOAL REACHED\n")

      for step, p in enumerate(path):
         print("Step", step)
         print(p[0:3])
         print(p[3:6])
         print(p[6:9])
         print()

      break

   i = s.index(0)

   for move in [-3, 3, -1, 1]:
      j = i + move

      if j < 0 or j > 8:
         continue

      if move == -1 and i % 3 == 0:
```

```
            continue

        if move == 1 and i % 3 == 2:
            continue

        temp = list(s)
        temp[i], temp[j] = temp[j], temp[i]
        temp = tuple(temp)

        if temp not in visited:
            visited.append(temp)
            stack.append((temp, path + [temp]))
```